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C# Application for Basic Arithmetic Operations.

Project type: C# Console Application

Steps to perform

1. Open Visual Studio and choose Create a new project.

2. Select Console App (C#), choose a project name and create it.

3. Open Program.cs and replace the sample code with the provided C# code.

4. Build the project; resolve any missing using directives or compilation errors.

5. Press Ctrl+F5, enter the required inputs, and verify the console output.

Closest supplied practical: Create an MVC application to perform basic Arithmetic Operations.

Source: Dot_Net_Practicals_1_to_5_.pdf

Provided creation instructions and source files

```
Creation steps: Create the MVC Application named ArthimeticMVC. Add the listed files at the
locations below and enter
the corresponding code.
Filename: Index.cshtml    Location: Views/ (MVC view folder)
@model ArithmeticMVC.Models.ArithmeticModel
@{
ViewBag.Title = "Arithmetic Operations";
}
<h2>Arithmetic Operation</h2>
@using (Html.BeginForm())
{
<table>
    <tr>
        <td>Frist Number</td>
        <td>@Html.TextBoxFor(m=>m.Number1)</td>
    </tr>
    <tr>
        <td>Second Number</td>
        <td>@Html.TextBoxFor(m=>m.Number2)</td>
    </tr>
    <tr>
        <td>Operation</td>
        <td>
            @Html.DropDownListFor(m=>m.Operation,
            new SelectList(new[]
            {
            "Add",
            "Subtract",
            "Multiply",
            "Divide"
            }))
        </td>
    </tr>
    <tr>
        <td></td>
        <td>
            <input type="submit" value="Calculate"/>
        </td>
    </tr>
</table>
}
@if(!string.IsNullOrEmpty(Model.ErrorMessage))
{
<h3 style="color:red">@Model.ErrorMessage</h3>
}
else if(Model.Operation!=null)
{
<h3>Result=@Model.Result</h3>
}
Filename: HomeController.cs    Location: Controllers/HomeController.cs
public class HomeController : Controller
{
    public ActionResult Index(ArithmeticModel model)
    {
        if (ModelState.IsValid)
        {
```

```
            switch (model.Operation)
            {
            case "Add":
                model.Result = model.Number1 + model.Number2;
                break;
            case "subtract":
                model.Result = model.Number1 - model.Number2;
                break;
            case "Multiply":
                model.Result = model.Number1 * model.Number2;
                break;
            case "Divide":
                if (model.Number2 == 0)
                {
                    model.ErrorMessage = "Divison by zero is not allowed.";
                }
                else
                {
                    model.Result = model.Number1 / model.Number2;
                }
                break;
            }
        }
        return View(model);
    }
}
Filename: ArithmeticModel.cs    Location: Models/ArithmeticModel.cs
using System.ComponentModel.DataAnnotations;
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
namespace ArithmeticMVC.Models
{
    public class ArithmeticModel
    {
        public double N
umber1 { get; set; }

        public double Number2 { get; set; }
        public string Operation { get; set; }
        public double Result { get; set; }
        public string ErrorMessage { get; set; }
    }
}
```

Expected result / verification

The application should meet the exact question requirements. Record actual output or screenshots after running; no execution has been claimed in this manual.